TRINITY — STATUS

Remnant Fieldworks Inc. · Coherent Inheritance Framework (CIF) · ExecutionProof-governed

TRINITY-2 — The Governed Action — HARDWARE COMPLETE, INDEPENDENTLY VERIFIED

As of 2026-07-19: executed on **three live IBM Heron r2 processors** after Derek rotated the IBM token and authorized the run. Manifest lock (`MANIFEST-TRINITY-2.sha256`) verified intact before execution. Aggregate verdict **ALLOW** → governed action **AUTHORIZED**; independent verifier **PASS**.

TRINITY-2 extends TRINITY-1 by binding the fused three-processor verdict to a **simulated $500,000 payment authorization**, with a preregistered **quorum (≥2 of 3) + substitution / failover** rule and full **calibration-snapshot hashing**. The payment is simulated — no real funds, rail, or authority.

Result (live IBM Heron r2 hardware, 2026-07-19)

Slot	Backend	S	Verdict	Job
D1	`ibm_kingston`	2.7650	PASS	`d9e7029ht-sac739dt2mg`
D2	`ibm_fez`	2.7235	PASS	`d9e71lqneu4c739o3otg`
D3	`ibm_marrakesh`	2.7235	PASS	`d9e71p-cjeosc73fie1r0`

- Shots: 4 CHSH circuits × 4,000 × 3 devices = **48,000 total**. Classical bound $|S| \le 2.0$ · Tsirelson $2\sqrt{2} \approx 2.828$.
- Quorum: required **2** of **3** · valid devices **3** · degraded_quorum **False**.
- Aggregate ExecutionProof verdict: **ALLOW** → governed action **AUTHORIZED** · `trinity_certified = True` .
- Fused nonce: `aa527fe2bb3b6d80e3ec78ab1558521a7b744f94235d805623129acf468602b2`
- Record hash: `4f6fa000e2bf3976a1d575267ecf188baf42f1c6caea9e717bf1fc1be6737dd0`
- Chain link → TRINITY-1 record hash `8d23accac81791fa10f1dba1be79a132168966c4620fc42d16de656bcf9d688b` .
- **Independent verifier (`trinity2_verify.py`): PASS** — every S, arm verdict, `device_witness_hash` , `calibration_snapshot_hash` , quorum, aggregate verdict, the governed-action decision, `governed_action_hash` , `fused_nonce` , and `record_hash` reconstructed from raw counts with no qiskit dependency; NIST + chain-link provenance OK.

PUBLISHED: GitHub `derekhone/trinity-testbeds` + Zenodo — concept DOI **10.5281/zenodo.21438244** (cite this), version v1 DOI 10.5281/zenodo.21438245 (2026-07-19).

Simulator validation (retained record — all branches confirmed pre-hardware)

Simulator validation (three independent noisy aer backends) — all branches confirmed + independently verified:

Scenario	Devices	Aggregate verdict	Action decision	Verifier
NOMINAL	3× PASS	**ALLOW**	AUTHORIZED	PASS
DEGRADED	2× PASS (1 device down)	**HOLD**	HELD_FOR_HUMAN_CONFIRMATION	PASS
DENY	2 PASS + 1 dissenting FAIL	**DENY**	DENIED	PASS
GATE-STOP	1 device (sub-quorum)	**GATE-STOP**	DENIED	PASS
TAMPER	flip a raw count post-hoc	(recompute)	—	**FAIL as required** (record_hash mismatch caught)
TAMPER	force decision AUTHORIZED under DENY	(recompute)	—	**FAIL as required**

- Kill-condition self-check active: harness exits non-zero if it ever emits ALLOW while a device is not PASS, under degraded quorum, or with fewer than 3 distinct devices.
- Independent verifier reconstructs every S, arm verdict, `device_witness_hash`, `calibration_snapshot_hash`, quorum, aggregate verdict, **the governed-action decision**, `governed_action_hash`, `fused_nonce`, and `record_hash` from raw counts — no qiskit.
- Schema `trinity-proofrecord-2.0`. Chain link → TRINITY-1 record hash `8d23accac81791fa10f1dba1be79a132168966c4620fc42d16de656bcf9d688b`.

Next (needs Derek): approve → rotate IBM token → run `python3 trinity2_harness.py --authorize-hardware --allow-unrotated` with `IBM_QUANTUM_TOKEN` set → verify → publish.

TRINITY-1 — The Three-Processor Witness — HARDWARE COMPLETE

As of: hardware run **COMPLETE**, independently verified, aggregate verdict **ALLOW**.

Result (live IBM Heron r2 hardware)

One preregistered Bell-CHSH nonclassicality witness, evaluated independently on three distinct quantum processors and fused into a single chain-linked ExecutionProof record.

Slot	Backend	S	nσ	Verdict	Job
D1	`ibm_kingston`	2.7360	23.27	VALID_ABOVE	`d9e46l-cjeosc73fiaj00`
D2	`ibm_fez`	2.5785	18.29	VALID_ABOVE	`d9e46p-cjeosc73fiaj50`
D3	`ibm_marrakesh`	2.7410	23.43	VALID_ABOVE	`d9e4a5sin-v1c73apq1pg`

- Classical bound $|S| \leq 2.0$ · Tsirelson $2\sqrt{2} \approx 2.828$ · certification threshold $S \geq 2.2$.
- Aggregate ExecutionProof verdict: **ALLOW** · `trinity_certified = True`
- TRINITY nonce: `508f0ad692a7bd3a1d1a069aa02f4dc2519560153f646220f2150bbe47270e2f`
- Record hash: `8d23accac81791fa10f1dba1be79a132168966c4620fc42d16de656bcf9d688b`
- Chain — previous record (OMNI-1): `22ffb2ba137c89c2846225c6d5a11fb2a4dc6aaa6a0ba1dfe6864b-c50393bb0c`
- NIST beacon pulse: `1865761` @ 2026-07-19T03:21:00.000Z
- Shots: 4,000 per circuit × 4 circuits × 3 devices = **48,000** total (matches prereg + locked harness `--shots` default; OMNI shot-count-consistency lesson honored).
- **Independent verifier: PASS** (recomputes every S, verdict, witness hash, nonce, record hash, and the aggregate verdict from raw counts — no qiskit).

Completed

- [x] **Preregistration** (`TRINITY-1-preregistration.md`) — three-processor design, identical CHSH witness, per-device + aggregate verdict logic (ALLOW/HOLD/DENY/GATE-STOP), fused `trinity_nonce` , kill condition, hardware gate, scope/honesty caveats, purpose gate.
- [x] **Harness** (`trinity_harness.py`) — builds the CHSH witness, runs three independent devices (simulator or hardware), assembles the `trinity-proofrecord-1.0` record, computes the aggregate ExecutionProof verdict, writes report + JSON. Records the **actual physical qubits** used after transpilation per circuit (post-layout).
- [x] **Independent verifier** (`trinity_verify.py`) — reconstructs every device S, verdict, `device_witness_hash` , `trinity_nonce` , and `record_hash` from raw counts, no qiskit.
- [x] **Preregistration + harness lock** (`MANIFEST.sha256`) — SHA-256, re-verified `OK` .
- [x] **Simulator validation** — three independent noisy aer backends:
- **NOMINAL** → all three VALID_ABOVE → **ALLOW**; verifier **PASS**.
- **HOLD** scenario → device 3 driven to INCONCLUSIVE → aggregate **HOLD**.
- **DENY** scenario → device 3 driven below classical (INVALID) → aggregate **DENY**.
- **Tamper test** → altering one raw count without recomputing the hash → verifier recomputes **DENY** and reports **FAIL** (record_hash mismatch caught).

- **Manifest-lock test** → editing the harness after locking broke the declared hash → provenance invalid → **DENY** (the lock works).
- [x] **Hardware run** on `ibm_kingston` + `ibm_fez` + `ibm_marrakesh` — genuine CHSH violation on all three, aggregate **ALLOW**, verifier **PASS**.

Scope & honesty (unchanged from preregistration)

- This is a **device-dependent** demonstration of nonclassicality reproduced across three processors and cryptographically fused into one governance record. It is **not** device-independent, does **not** close the detection or locality loopholes, and does **not** entangle the three processors with one another. Each device is a separate, self-contained CHSH experiment; TRINITY is the reproduction + fusion claim, nothing more.

Provenance / reproducibility

```
cd /home/ubuntu/trinity-testbeds
sha256sum -c MANIFEST.sha256          # confirm prereg + harness lock intact
python3 trinity_verify.py             # independent reconstruction from raw counts →
PASS
```

Security note

The IBM Quantum token used for this run was pasted in chat and is therefore considered **compromised** — rotate it at https://quantum.cloud.ibm.com. The harness only reads the token from the `IBM_QUANTUM_TOKEN` environment variable — never hard-coded, never committed (`.gitignore` excludes `.env`).